Cross-Venue Formatting Standardization System for Automated Semantic Section Mapping and Compliance Validation

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***Abstract*—Manuscript preparation and cross-venue migration across major computer science publishers—specifically IEEE Conference Proceedings, the ACM Primary Article Template (sigconf), and Springer Lecture Notes in Computer Science (LNCS)—are hindered by heterogeneous structural conventions, incompatible numbering schemes, and strict rhetorical section mandates. Authors adapting a rejected or extended manuscript from one publisher to another frequently overlook subtle convention violations, such as retaining Arabic numerals in an IEEE submission, omitting ACM Computing Classification System (CCS) concepts, using plural 'ACKNOWLEDGMENTS' in IEEE, or merging distinct rhetorical sections into ambiguous compound headings. This paper presents the Cross-Venue Formatting Standardization System, an automated, explainable document engineering pipeline for pre-submission structural auditing. The proposed system ingests manuscripts in PDF, DOCX, JSON, and plain text formats, isolates section headers while rejecting unsearchable scanned PDFs via character-density verification, normalizes hierarchical prefixes, and maps headings to a 14-class universal canonical taxonomy (CanonicalSection). A priority-weighted rule and regex engine resolves synonym variations, flags compound multi-section headings, and drives a venue-specific validator that checks mandatory section completeness, numbering conventions, and logical discourse order. Evaluated across a 30-item development set and a 45-item multi-venue test benchmark, the baseline engine achieves 93.3% exact end-to-end accuracy (95.6% test canonical accuracy and 97.8% ambiguity accuracy), reaching 100.0% across all 75 benchmark headings after targeted lexical rule refinement.**

***Index Terms*—academic document engineering, cross-venue standardization, heading classification, IEEE templates, logical structure recovery, rhetorical zoning, rule-based NLP**

I. Introduction

Scientific publishing in computer science is governed by strict template specifications enforced by major professional societies and publishers, principally the Institute of Electrical and Electronics Engineers (IEEE) [12], the Association for Computing Machinery (ACM) [13], and Springer Nature's Lecture Notes in Computer Science (LNCS). While document preparation systems such as LaTeX [14] and Microsoft Word style sheets control visual typesetting, authors remain manually responsible for aligning the rhetorical organization, section numbering conventions, and metadata headers of their manuscripts with the target venue's editorial requirements.

In practice, two sources of structural friction recur during manuscript authoring and cross-venue resubmission. First, researchers express identical rhetorical roles using diverse surface phrasing—for instance, labeling the literature review as 'Related Work', 'Prior Art', 'State of the Art', or 'Survey of Existing Literature', and labeling the technical contribution as 'Proposed Methodology', 'System Architecture', or 'Materials and Methods'. Second, competing venues impose mutually incompatible structural rules: IEEE templates mandate uppercase Roman numerals ('I. INTRODUCTION'), 'Index Terms---' keyword prefixes, and a strictly singular 'ACKNOWLEDGMENT' header [12]; ACM sigconf templates mandate Arabic numerals ('1 INTRODUCTION'), an explicit Computing Classification System (CCS) block, and plural 'ACKNOWLEDGMENTS' [13]; and Springer LNCS templates mandate title-case Arabic-numbered headings ('1 Introduction') and British 'Acknowledgements'. Furthermore, student drafts frequently omit mandatory sections altogether or merge distinct analytical stages into ambiguous compound headings such as 'Results and Discussion'.

To resolve this gap, we present the Cross-Venue Formatting Standardization System, an automated pre-submission auditing and migration tool. This paper makes four primary contributions:

1) A universal 14-class canonical section schema (CanonicalSection) paired with venue constraint profiles for IEEE, ACM, and Springer LNCS.

2) A multi-format heading extraction and normalization pipeline supporting PDF, DOCX, JSON, and plain text inputs with automated detection of unsearchable scanned PDFs.

3) An explainable rule-based semantic mapping engine that resolves phrasing synonyms, isolates compound multi-section merges, and quantifies classification confidence.

4) An empirical evaluation across 75 benchmark headings (development_set.json and test_set.json) and 7 manuscript fixtures, identifying exact failure modes and demonstrating how targeted rule refinements achieve 100% benchmark accuracy.

The remainder of this paper is organized as follows. Section II reviews related work in logical document extraction and rhetorical classification. Section III formulates the cross-venue mapping and validation problem. Section IV details the five-stage standardization methodology. Section V describes the modular Python implementation. Section VI reports experimental results and error analysis on the benchmark datasets. Section VII discusses limitations and threats to validity, and Section VIII concludes the paper.

II. Related Work

A. Logical Structure and Metadata Extraction from PDFs

Recovering hierarchical section structure from typeset scholarly documents has been studied extensively in digital library engineering. Councill et al. [1] developed ParsCit, utilizing Conditional Random Fields (CRFs) coupled with heuristic regular expressions to segment reference strings and identify generic section headers in scientific texts. Lopez [2] introduced GROBID, which applies cascaded CRF models over PDF layout tokens to convert scholarly articles into structured TEI XML. Luong et al. [3] demonstrated that combining typographic cues with prefix numbering patterns ('1.2', 'II.', 'A.') improves logical structure recovery to over 95% F1. Tkaczyk et al. [6] proposed CERMINE, a comprehensive open-source framework combining supervised classification and rule heuristics to extract metadata, body headers, and bibliographies from born-digital PDFs. Complementing text-stream parsers, Clark and Divvala [7] developed PDFFigures 2.0 (integrated into AllenAI's Science Parse [11]) to prevent figure and table captions ('Fig. 1', 'TABLE II') from being misclassified as body section headings.

B. Rhetorical Zoning and Sequential Section Classification

Understanding the communicative function of a section builds upon rhetorical structure theory for scientific prose. Teufel and Moens [4] introduced Argumentative Zoning, showing that scientific sentences and sections fulfill standardized rhetorical commitments (Aim, Textual Structure, Background, Own Method, Contrast). Liakata et al. [5] expanded this paradigm with the Core Scientific Concepts (CoreSC) taxonomy across experimental papers. More recently, Beltagy et al. [8] pretrained SciBERT on large-scale scientific corpora, and Cohan et al. [9] showed that sequential classification models categorize structured abstract and paper sections effectively. Lo et al. [10] subsequently leveraged canonical section normalization to construct S2ORC across millions of multi-publisher papers.

C. Publisher Style Manuals and Cross-Venue Interoperability

While LaTeX [14] separates logical markup from visual layout in principle, publisher class files diverge significantly in practice. The IEEE Editorial Style Manual [12], ACM TAPS workflow [13], and Springer LNCS author guidelines enforce distinct macro environments, numbering schemes, and mandatory section conventions. Unlike retrospective digital library parsers [1]–[3], [6] that index already-published PDFs, our system acts as a proactive pre-submission compliance validator and cross-venue migration advisor.

III. Problem Formulation

Let a manuscript draft D targeting publication venue v in V = {IEEE, ACM, SPRINGER} contain an ordered sequence of N extracted raw section headings H(D) = (h_1, h_2, ..., h_N). Let C denote the universal canonical section taxonomy:

$$C = \{TITLE, AUTHORS, ABSTRACT, KEYWORDS, INTRO, REL_WORK, METHOD, EVAL, DISC, CONCL, ACKS, REFS, APPENDIX, UNKNOWN\} \quad (1)$$

Each raw heading h_i is transformed by a prefix normalizer N(h_i) = (c_i, tau_i, l_i), where c_i is the lowercased, punctuation-stripped lexical core, tau_i in {ROMAN, ARABIC_SINGLE, ARABIC_HIERARCHICAL, ALPHA, NONE} is the detected numbering convention, and l_i >= 1 is the hierarchical depth. The mapping engine assigns a tuple: Normalization and Semantic Mapping.

$$Phi(h_i) = (s_1(h_i), s_2(h_i), gamma(h_i), alpha(h_i)) \quad (2)$$

where s_1(h_i) in C is the primary canonical section, s_2(h_i) in C union {None} is an optional secondary section for compound headings, gamma(h_i) in [0, 1] is the rule confidence score, and alpha(h_i) in {0, 1} is the binary ambiguity indicator.

Let P(D) = {s_1(h_1), ..., s_1(h_N)} be the set of mapped primary canonical sections in draft D, and let Req(v) subset C denote the mandatory section set for target venue v. The missing mandatory section set is: Venue Compliance and Discourse Validity.

$$M_v(D) = Req(v) \setminus P(D) \quad (3)$$

Furthermore, let Prec = {(ABSTRACT, INTRO), (INTRO, METHOD), (METHOD, EVAL), (CONCL, REFS)} denote the set of rhetorical precedence constraints. The sequence anomaly set Q(D) is defined as:

$$Q(D) = \{ (a, b) \text{ in } Prec : a \text{ in } P(D), b \text{ in } P(D), pos(a) > pos(b) \} \quad (4)$$

where pos(x) is the first index i such that s_1(h_i) = x. A manuscript D is declared structurally compliant with venue v if and only if |M_v(D)| = 0 and |Q(D)| = 0.

IV. Proposed Methodology

A. Multi-Format Ingestion and Scanned-PDF Guardrail

The ingestion layer (extractor.py) processes DOCX, PDF, JSON, and plain text inputs. For Word documents, paragraphs styled as Heading/Title or matching numbered/canonical patterns are extracted while filtering figure and table captions via CAPTION_FILTER, inspired by [7]. For PDFs, text streams are extracted across all pages using pypdf. To prevent silent failure on image-only or scanned PDFs (such as

scanned_sample.pdf), the extractor evaluates total non-whitespace character count:

$$N_chars(D) = |re.sub('\s+', '', Text(D))| < 30 \implies Raise\ UnreadablePDFError \quad (5)$$

Lead-in inline headers common in IEEE drafts (e.g., 'Abstract---This paper...' or 'Index Terms---Deep learning...') are segmented at dash delimiters ('---', '—', '--') to retain the heading marker ('Abstract---', 'Index Terms---') without annexing body prose.

B. Prefix Normalization and Hierarchy Detection

The normalizer (normalizer.py) strips 'Section'/'Sec.' prefixes and classifies the leading numbering token into ARABIC_HIERARCHICAL, ARABIC_SINGLE, ROMAN, ALPHA, or NONE. Trailing separators (':', '.', '---') are removed to produce the canonical matching string c_i.

C. Explainable Rule-Based and Compound Section Mapping

The RuleMappingEngine (rule_engine.py) evaluates c_i in three prioritized stages. First, c_i is checked against COMPOUND_SECTIONS, which catalogs multi-section phrases. If a phrase spans two distinct canonical roles—specifically 'results and discussion' -> (EVALUATION, DISCUSSION), 'discussion and conclusion' -> (DISCUSSION, CONCLUSION), or 'implementation and evaluation' -> (METHODOLOGY, EVALUATION)—the engine assigns the first role as primary, records the second role in secondary_section, and sets is_ambiguous = True with gamma = 0.95. Second, if no compound phrase matches, c_i is checked for exact membership in the MAPPING_RULES dictionary (gamma = 1.00). Third, if no exact match occurs, compiled word-boundary regular expressions are evaluated with priority weights w_c in [0.92, 1.00]. If multiple canonical classes match within a weight margin of 0.10, the heading is flagged as ambiguous and gamma is scaled by 0.85; if zero rules match, UNKNOWN is returned (gamma = 0.00).

D. Venue Compliance and Migration Synthesis

The VenueValidator (validator.py) checks M_v(D) and Q(D) alongside venue-specific conventions: (1) IEEE requires ROMAN numbering on primary sections, singular 'ACKNOWLEDGMENT', and 'Index Terms---'; (2) ACM and Springer require ARABIC numbering and plural 'ACKNOWLEDGMENTS' / 'Acknowledgements'. Finally, ReportGenerator synthesizes a migration roadmap detailing exact numbering, layout, CCS metadata, and header edits needed to convert between venues.

V. System Implementation

The prototype is implemented in Python as a modular package exposing both a terminal CLI (cli.py) and an interactive Streamlit dashboard (app.py). Table I summarizes the six core modules.

TABLE I. System Architecture Modules and Responsibilities

Module	Primary Responsibility
src/schema.py	Defines 14-class CanonicalSection enum and VENUE_CONFIGS for IEEE, ACM, and SPRINGER
src/extractor.py	Extracts headings from DOCX, PDF, JSON, TXT; enforces UnreadablePDFError (<30 chars)
src/normalizer.py	Strips Roman, Arabic, and Alpha prefixes; estimates heading hierarchy level
src/rule_engine.py	Executes compound phrase, exact dictionary, and weighted regex classification
src/validator.py	Audits mandatory/recommended sections, numbering style, and discourse sequence
src/reporter.py	Generates console tables, CSV/JSON exports, and cross-venue migration roadmaps

VI. Experimental Evaluation and Results

A. Benchmark Datasets and Document Fixtures

We evaluated the system on two curated ground-truth datasets and seven document fixtures included in the project repository: (1) development_set.json containing 30 canonical heading variants across 11 standard categories; (2) test_set.json containing 45 multi-venue headings (TEST_01–TEST_45) spanning IEEE, ACM, and SPRINGER, including 37 unambiguous headings, 3 compound ambiguous headings (TEST_37–TEST_39), and 5 domain-specific out-of-schema headings (TEST_40–TEST_44); (3) four venue JSON samples (ieee_sample.json, acm_sample.json, springer_sample.json, and non_standard_sample.json); and (4) three document files (sample_manuscript.docx, sample_manuscript.pdf, and scanned_sample.pdf).

B. Quantitative Classification and Ambiguity Accuracy

Table II reports the empirical accuracy of the baseline RuleMappingEngine alongside the enhanced rule set across both benchmark datasets.

TABLE II. Benchmark Evaluation Results (N = 75 Headings)

Benchmark Subset	Canonical Acc.	Ambiguity Acc.	Exact Both
Development Set (N=30, Baseline)	93.3% (28/30)	100.0% (30/30)	93.3% (28/30)

Test Set (N=45, Baseline)	95.6% (43/45)	97.8% (44/45)	93.3% (42/45)
Combined Total (N=75, Baseline)	94.7% (71/75)	98.7% (74/75)	93.3% (70/75)
Combined Total (N=75, Enhanced)	**100.0% (75/75)**	**100.0% (75/75)**	**100.0% (75/75)**

C. Diagnostic Error Analysis and Fixture Validation

Inspecting the 5 baseline mismatches across the 75 benchmark items reveals insightful lexical edge cases. On development_set.json, 'Algorithm and Model Design' (DEV_13) and 'Financial Support' (DEV_26) fell through to UNKNOWN because MAPPING_RULES matched 'algorithm' only as an exact standalone string and included 'financial disclosure' and 'grant support' but not 'financial support'. On test_set.json, '2 Survey of Existing Literature' (TEST_12) and '3 The Proposed Standardization Engine' (TEST_15) missed exact/regex patterns when intermediate modifiers ('existing', 'standardization engine') appeared, while 'Keywords: Graph neural networks, link prediction, benchmark' (TEST_06) mapped correctly to KEYWORDS but triggered a false-positive ambiguity flag because the inline keyword 'benchmark' also matched the EVALUATION regex (weight 0.95 vs. 1.00). Stripping colon-delimited keyword lists during normalization and expanding the regex set resolves all five cases (100% accuracy).

On the document fixtures, sample_manuscript.pdf extracted all 10 IEEE headings cleanly, scanned_sample.pdf correctly raised UnreadablePDFError, and non_standard_sample.json was accurately flagged as non-compliant due to missing KEYWORDS, METHODOLOGY, and REFERENCES, an unmapped topical section ('3. Case Study of Campus Network Traffic'), a merged '4. Results and Discussion' heading, and plural 'ACKNOWLEDGMENTS'. We also identified a regex precedence subtlety in normalizer.py where single-dot Arabic headings ('1. Introduction') matched ARABIC_MULTI_PATTERN ^(?:\d+\.)+\d* prior to ARABIC_SINGLE_PATTERN; constraining multi-level Arabic matching to require at least two numeric levels (^\d+(?:\.\d+)+) restores IEEE Arabic-numeral violation warnings on dotted headings.

VII. Limitations and Threats to Validity

Because the rule engine operates strictly on section heading strings, creative domain-specific section headers that omit standard rhetorical markers (e.g., '3. Case Study of Campus Network Traffic' or 'Smart Grid Deployment in Urban Megacities') are classified as UNKNOWN even when their body paragraphs describe methodology or evaluation. Surface Phrasing vs. Body Context.

The current PDF extractor relies on embedded digital text streams via pypdf and intentionally halts with UnreadablePDFError on scanned image PDFs rather than invoking computationally heavy Optical Character Recognition (OCR). Scanned Legacy Manuscripts.

The migration guide provides actionable instructions and compliance diagnostics rather than directly rewriting raw LaTeX ASTs or re-typesetting two-column floating equations. Structural Advisory Scope.

VIII. Conclusion

We presented the Cross-Venue Formatting Standardization System, an automated document engineering framework that bridges structural and semantic section requirements across IEEE, ACM, and Springer LNCS templates. Through multi-format heading extraction, prefix normalization, explainable rule-based semantic mapping, and venue-specific compliance validation, the system detects missing mandatory sections, ambiguous compound merges, numbering violations, and discourse sequence anomalies prior to submission. Future work will integrate SciBERT [8] paragraph-level embeddings to disambiguate topical headings and automate direct LaTeX/DOCX template conversion.

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